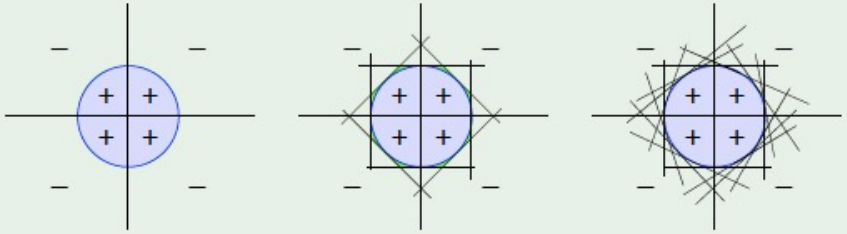
CSE4088 Introduction to Machine Learning

Overfitting

Slides are adopted from lecture notes of Yaser Abu-Mostafa

Review of last lecture

- Multilayer perceptrons

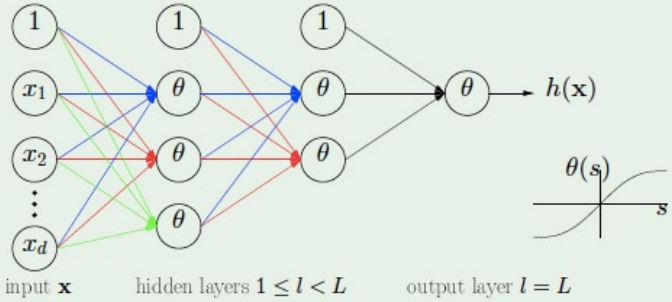


Logical combinations of perceptrons

- Neural networks

$$x_j^{(l)} = \theta\left(\sum_{i=0}^{d^{(l-1)}} w_{ij}^{(l)}\, x_i^{(l-1)}\right)$$

where $\theta(s) = \tanh(s)$



- Backpropagation

$$\Delta w_{ij}^{(l)} = -\eta\, x_i^{(l-1)} \delta_j^{(l)}$$

where

$$\delta_i^{(l-1)} = (1 - (x_i^{(l-1)})^2) \sum_{j=1}^{d^{(l)}} w_{ij}^{(l)}\, \delta_j^{(l)}$$

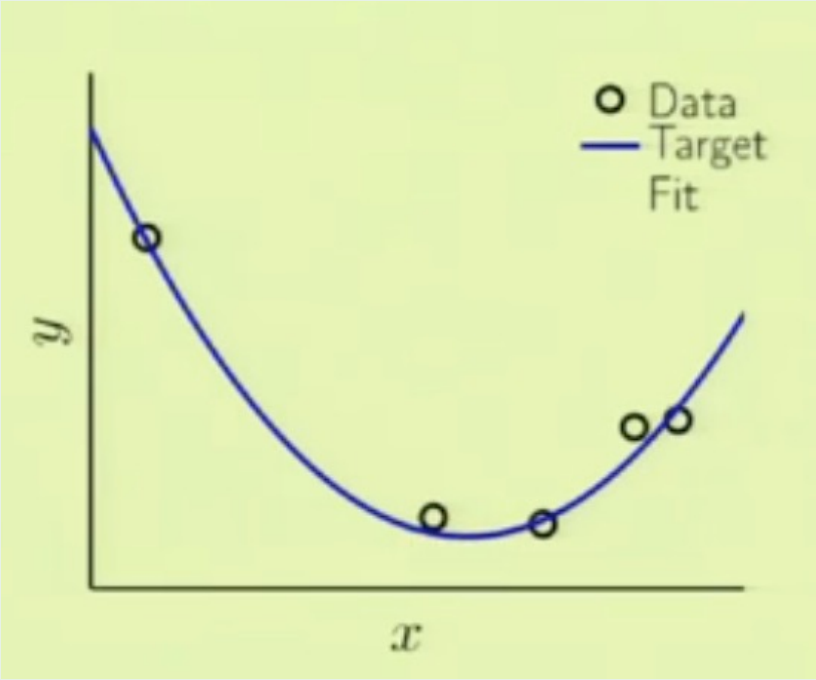
Outline

- What is overfitting?
- The role of noise
- Deterministic noise
- Dealing with overfitting

Illustration of overfitting

Simple target function

5 data points- **noisy**



Black points have been sampled from the blue target with little noise.

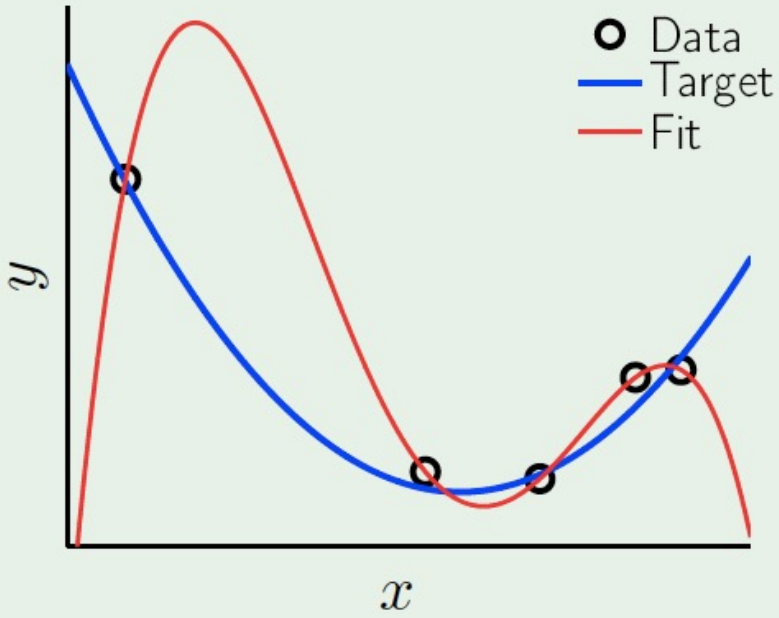
Illustration of overfitting

Simple target function

5 data points- **noisy**

4th-order polynomial fit

$$E_{\text{in}} = 0, \quad E_{\text{out}} \text{ is huge}$$



We fit a 4th order polynomial to the five black points.

Overfitting can occur if you increase the model order more than you should.

Overfitting versus bad generalization

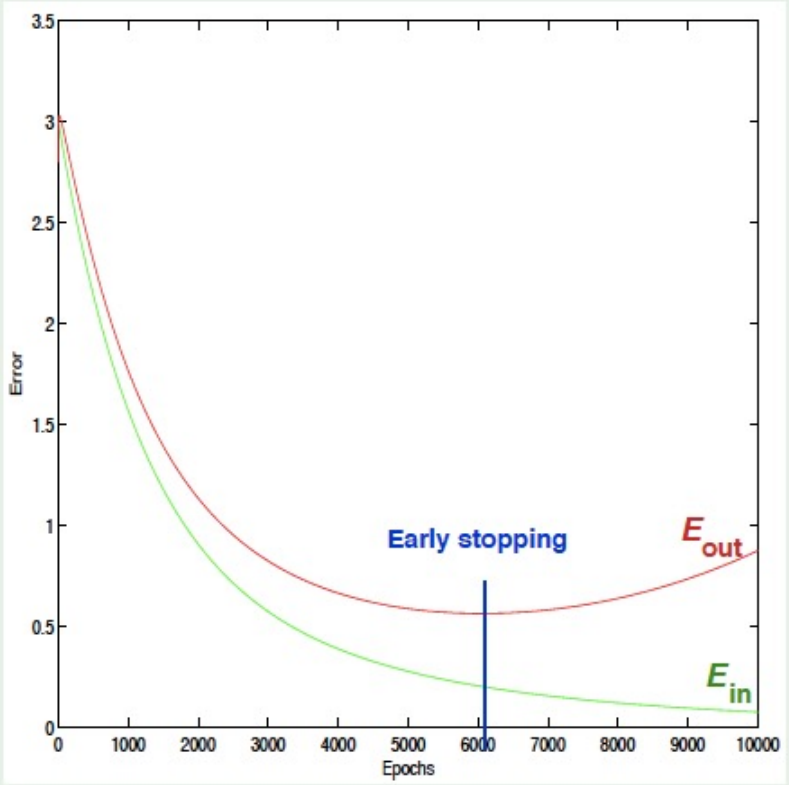
Neural network fitting noisy data

Overfitting: $E_{\text{in}} \downarrow$ $E_{\text{out}} \uparrow$

Overfitting can occur within the same model (e.g. neural network).

The generalization error is the distance between the red and green curve.

Overfitting starts around the blue line. Since E_{out} starts increasing from then on.



The culprit (source of the problem)

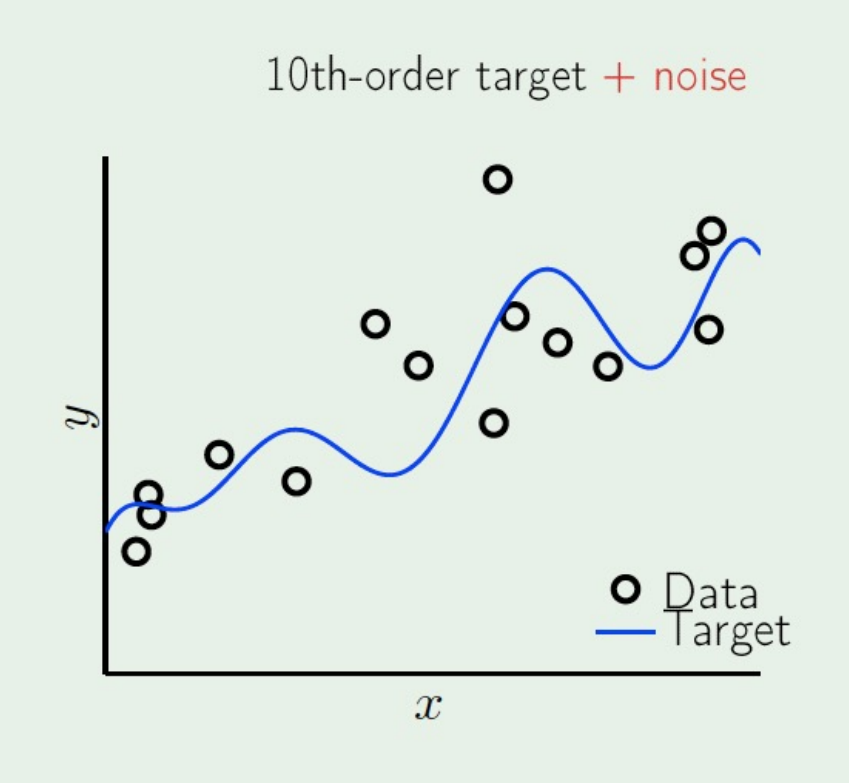
Overfitting: "fitting the data more than is warranted"

Culprit: fitting the noise - harmful

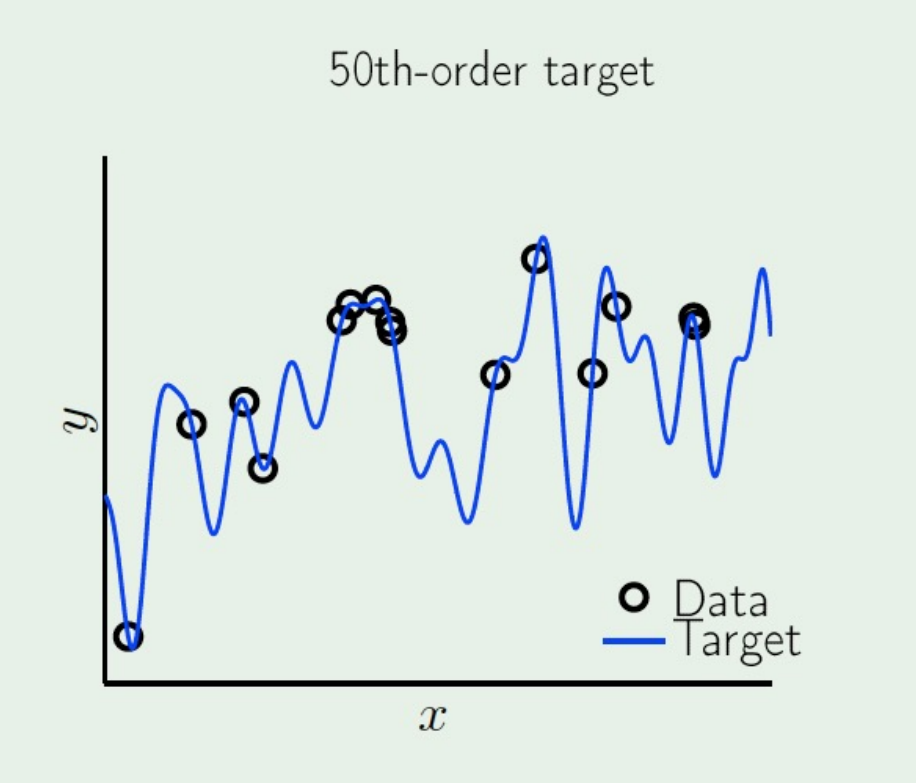
When there is overfitting, we are fitting the noise and the model is hallucinating a pattern in the noise, which becomes harmful later on.

When you are trying to get the sample right, you are inadvertently fitting the noise. There is no pattern in the noise, hence we can not learn the noise. Fitting the noise is harmful.

Case study

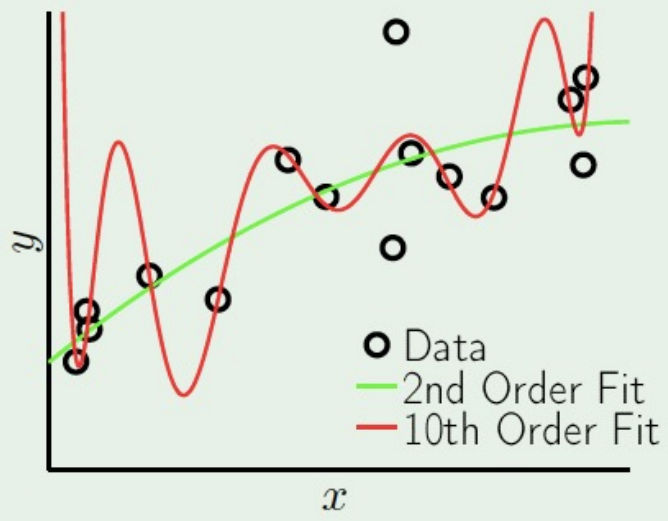


Generated 15 black points **with noise** from the blue target function (10th order).



Generated 15 black points **without noise** from the complex blue target (50th order).

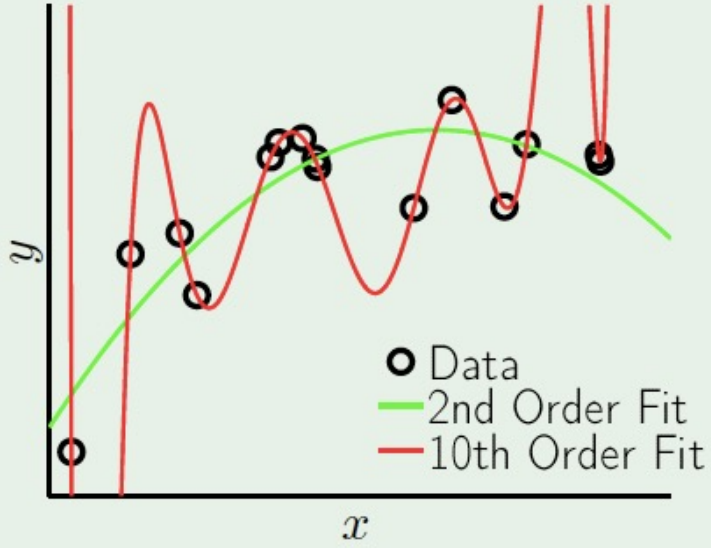
Two fits for each target



Noisy low-order target (10th order)

	2nd Order	10th Order
E_{in}	0.050	0.034
E_{out}	0.127	9.00

We are overfitting with the 10th order polynomial.
E_{in} decreased but E_{out} increased. Fitting the noise.



Noiseless high-order target (50th order)

	2nd Order	10th Order
E_{in}	0.029	10^{-5}
E_{out}	0.120	7680

We are overfitting with the 10th order polynomial.
E_{in} decreased by E_{out} increased!! But there is no noise here? Fitting another noise

An irony of two learners

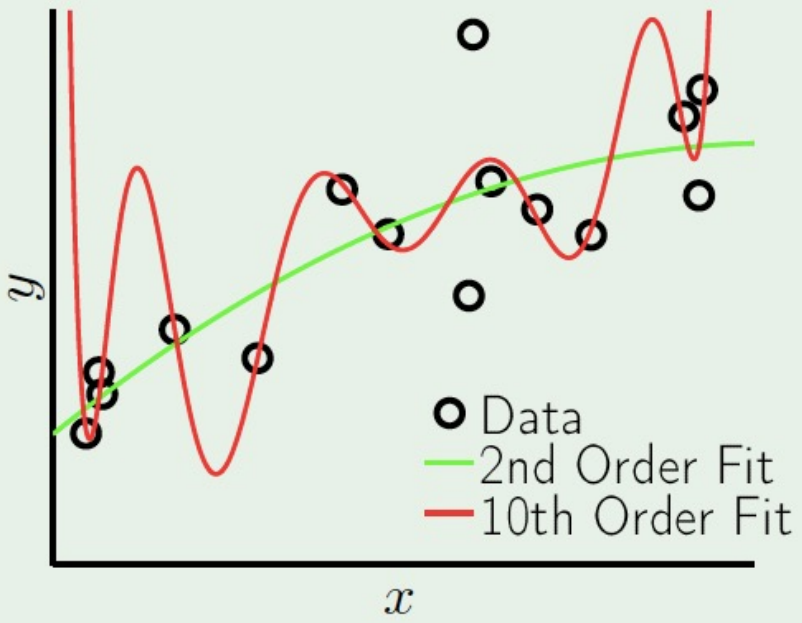
Two learners O and R

They know the target is 10th order

They have 15 **noisy samples**. They will choose a model.

O chooses $\mathcal{H}_{10}$ R chooses $\mathcal{H}_2$

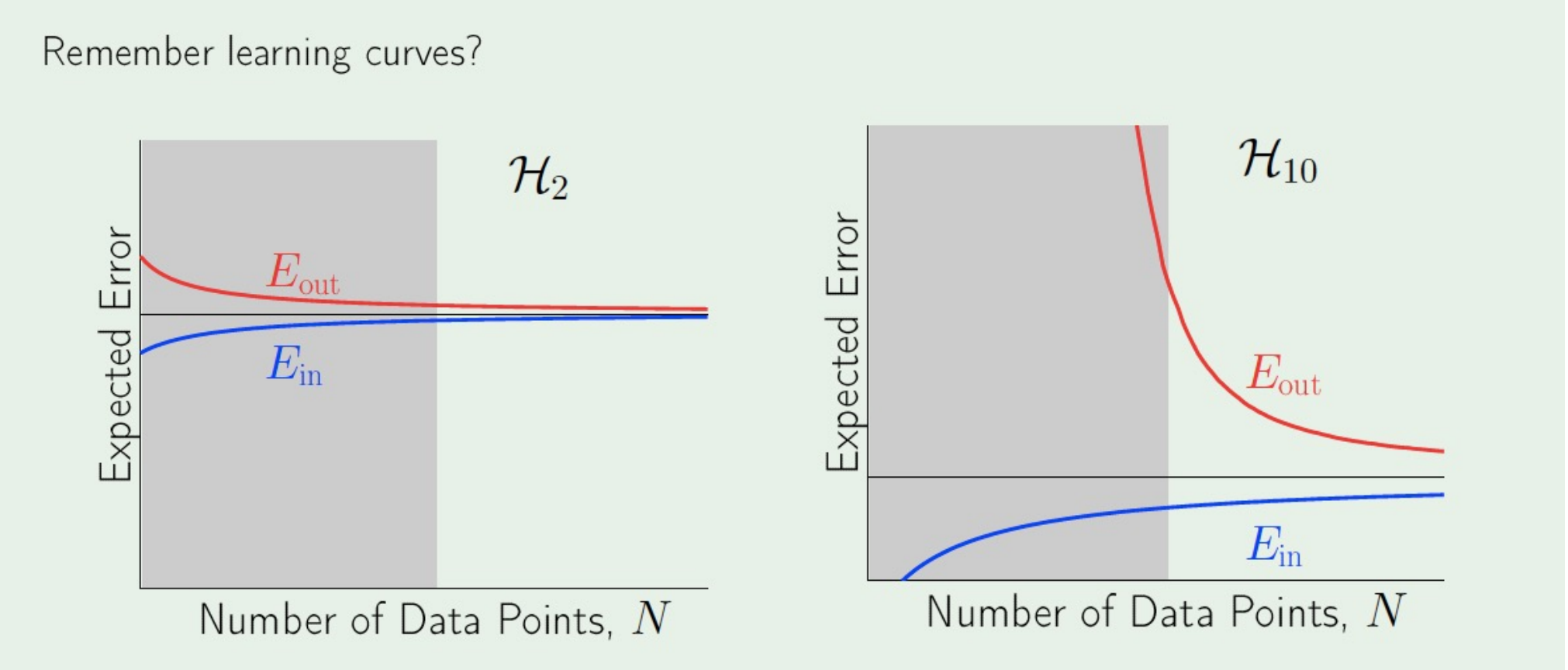
There are 15 points (N).
H_2 has 3 parameters, ratio = 15/3 = 5.
Rule of thumb says ratio should be at least 10.
H_2 is more reasonable to learn as compared to H_{10}



Learning a 10th-order target

Remember: You must match the complexity of your hypothesis to the data resources, not the complexity of the target function.

We have seen this case



The generalization error (difference between E_in and E_out) is small. Inevitable error comes from the limitation of the model and the noise. Inevitable error is large. As the number of N increases, E_in and E_out always becomes closer.

If N is small, there is overfitting (gray area), since E_out is very large E_in is small. When N increases, they converge to a smaller error since the hypothesis has the capacity to model the target. The error is due to the noise.
H_{10} lost because of the insufficient number of samples. If N is sufficient, this model will win.

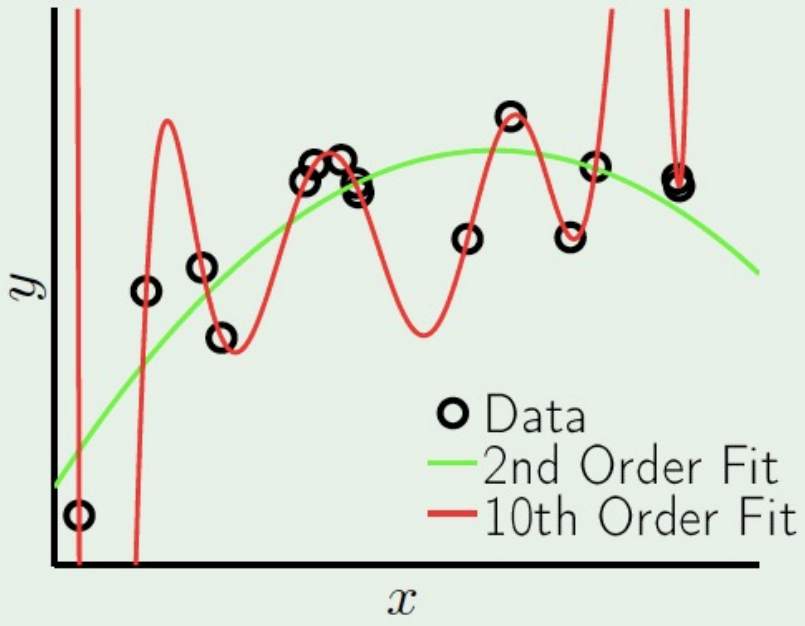
Even without noise

The two learners $\mathcal{H}_{10}$ and $\mathcal{H}_2$

They know there is no noise

Is there really no noise?

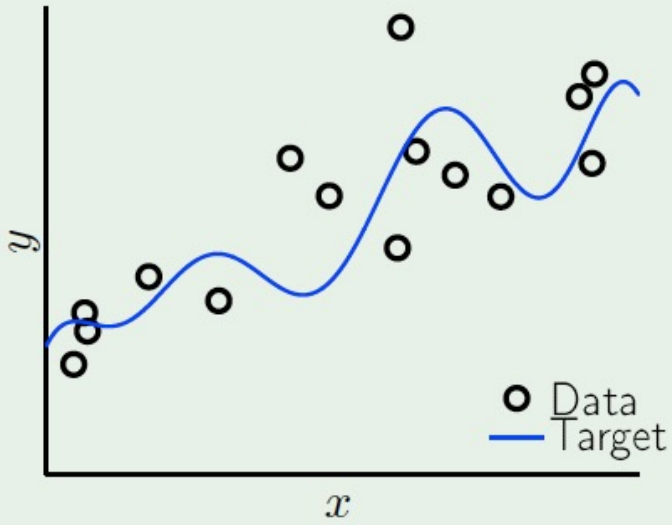
H_{10} has a better chance of getting closer to the target of 50th order polynomial. But it has larger $E_{out.}$ Why?



Learning a 50th-order target

A detailed experiment

Impact of **noise level** and **target complexity**



$$y = f(x) + \underbrace{\epsilon(x)}_{\sigma^2} = \underbrace{\sum_{q=0}^{Q_f} \alpha_q \, x^q}_{\text{normalized to energy 1}} + \epsilon(x)$$

noise level: σ^2

The polynomial coefficents are normalized so that they add up to one. If they get too large, signal to noise ratio will be too high and noise will have no effect.

target complexity: Q_f

data set size: N

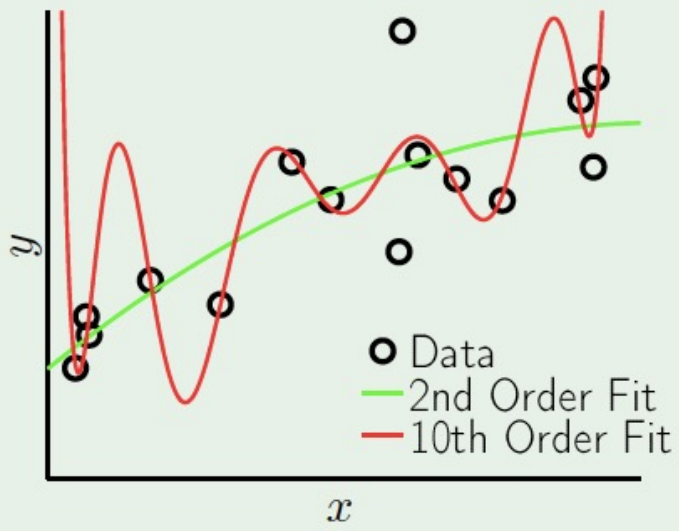
If we are given more data we are less susceptibel to overfitting.

One instance: $Q_f = 10$, $N = 15$

The overfit measure

We fit the data set $(x_1, y_1), \cdots, (x_N, y_N)$ using our two models:

$\mathcal{H}_2$: 2nd-order polynomials $\mathcal{H}_{10}$: 10th-order polynomials



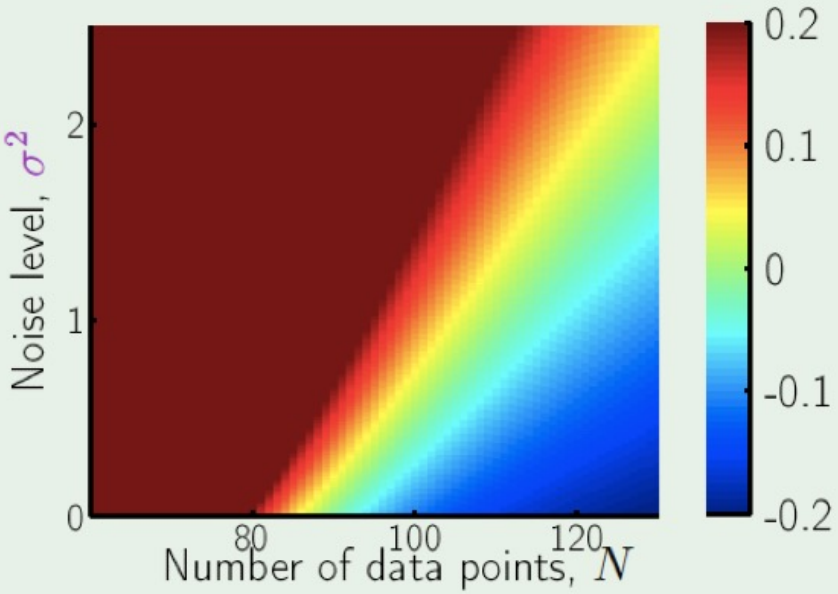
Compare out-of-sample errors of

$g_2 \in \mathcal{H}_2$ and $g_{10} \in \mathcal{H}_{10}$

overfit measure: $E_{\text{out}}(g_{10}) - E_{\text{out}}(g_2)$

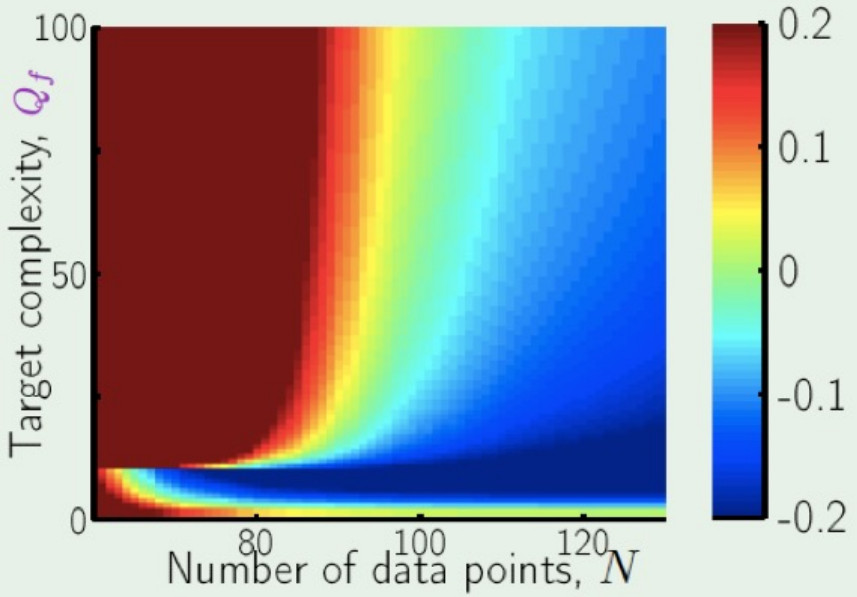
When the this difference is positive,
complex model is doing worse, we have an overfit.
Otherwise there is no overfitting.

The results



Impact of σ^2

$Q^f = 20$ (order of target polynomial is fixed)



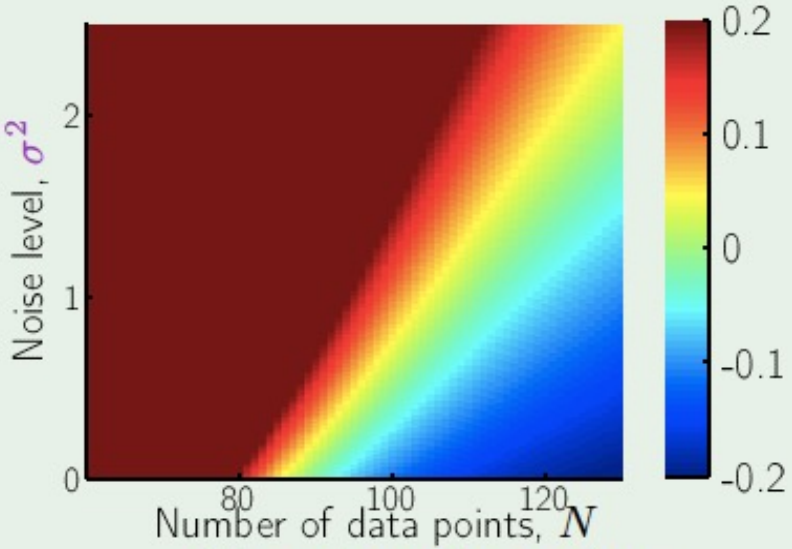
Impact of Q_f

Noise level $\sigma^2 = 1$ (fixed)

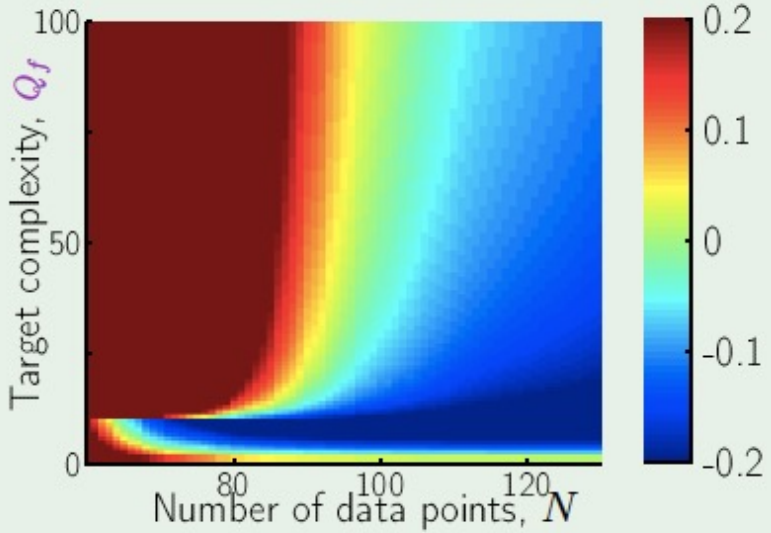
overfit measure: $E_{\text{out}}(g_{10}) - E_{\text{out}}(g_2)$

Red: more overfitting **Blue**: less overfitting

Impact of "noise"



Stochastic noise



Deterministic noise

number of data points	↑	Overfitting	↓
stochastic noise	↑	Overfitting	↑
deterministic noise	↑	Overfitting	↑

Outline

- What is overfitting?
- The role of noise
- Deterministic noise
- Dealing with overfitting

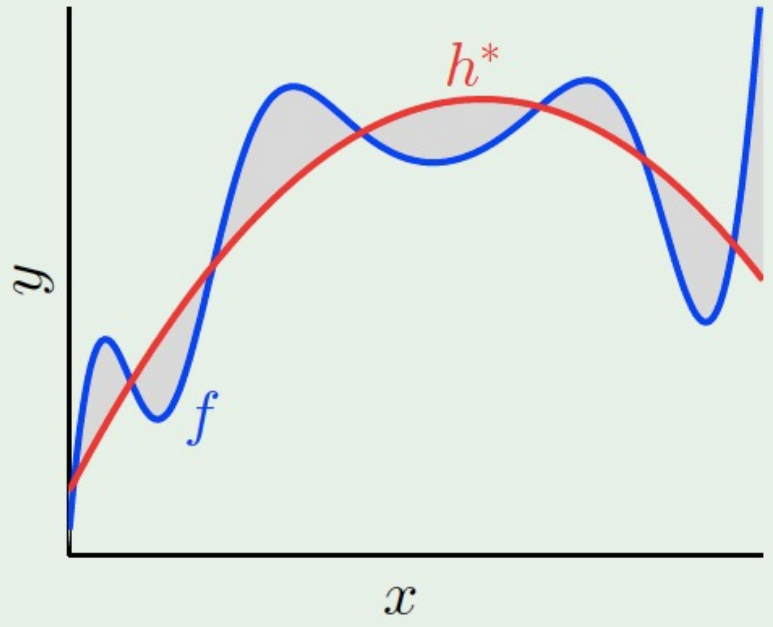
Definition of deterministic noise

The part of f that $\mathcal{H}$ cannot capture: $f(\mathbf{x}) - h^*(\mathbf{x})$

Why "noise"?

Main differences with stochastic noise:

1. depends on $\mathcal{H}$
2. fixed for a given $\mathbf{x}$



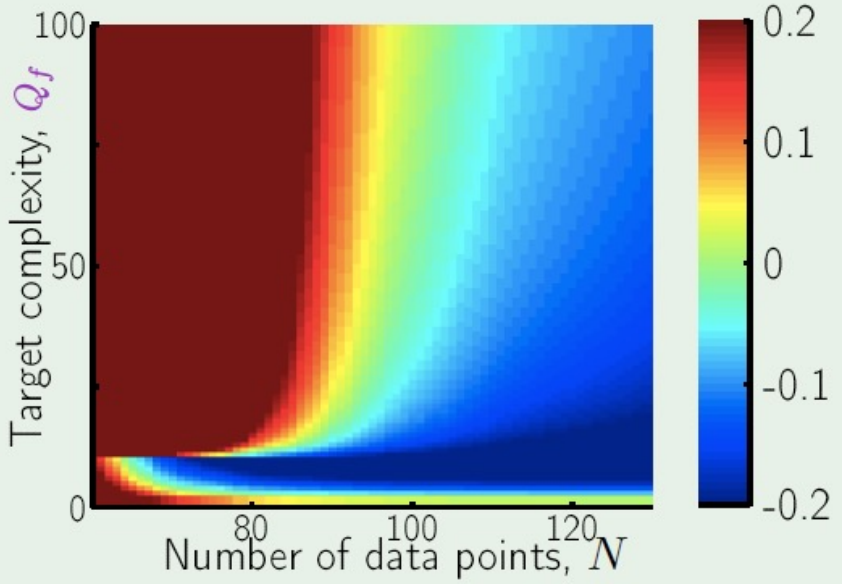
-- Deterministic noise depends on H: as the hypothesis set gets larger, the deterministic noise gets smaller.
-- Deterministic noise fixed at a certain x.

-- h* is the best possible approximation of *f* (target)
-- The shaded region is the deterministic noise.

Impact on overfitting

Deterministic noise and Q_f

Finite N: $\mathcal{H}$ tries to fit the noise



how much overfit

overfit measure: $E_{\text{out}}(g_{10}) - E_{\text{out}}(g_2)$

The bottom part of figure is interesting:
--We are looking at the difference between E_{out} of 10th orger polynomial minus the error of second order polynomial.
--There is a jump in the figure at Q_f = 10. Q should be larger that 10 so that there is something we cannot approximate.

Noise and bias-variance

Recall the decomposition:

$$\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x})\right)^2\right] = \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x})\right)^2\right]}_{\text{var}(\mathbf{x})} + \underbrace{\left[\left(\bar{g}(\mathbf{x}) - f(\mathbf{x})\right)^2\right]}_{\text{bias}(\mathbf{x})}$$

What if f is a noisy target?

$$y = f(\mathbf{x}) + \epsilon(\mathbf{x}) \qquad \mathbb{E}\left[\epsilon(\mathbf{x})\right] = 0$$

Expected value of out of sample error was decomposed into two terms.

A noise term

y is noisy

$$\mathbb{E}_{\mathcal{D},\epsilon}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \quad y \quad\right)^2\right] = \mathbb{E}_{\mathcal{D},\epsilon}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - f(\mathbf{x}) - \epsilon(\mathbf{x})\right)^2\right]$$

$$= \mathbb{E}_{\mathcal{D},\epsilon}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x}) + \bar{g}(\mathbf{x}) - f(\mathbf{x}) - \epsilon(\mathbf{x})\right)^2\right]$$

$$= \mathbb{E}_{\mathcal{D},\epsilon}\left[\left(g^{(\mathcal{D})}(\mathbf{x}) - \bar{g}(\mathbf{x})\right)^2 + \left(\bar{g}(\mathbf{x}) - f(\mathbf{x})\right)^2 + \left(\epsilon(\mathbf{x})\right)^2\right.$$

$$\left. + \text{ cross terms }\right]$$

Actually, two noise terms

$$\underbrace{\mathbb{E}_{\mathcal{D},\mathbf{x}}\left[\left(g^{(\mathcal{D})}(\mathbf{x})-\bar{g}(\mathbf{x})\right)^2\right]}_{\text{var}}+\underbrace{\mathbb{E}_{\mathbf{x}}\left[\left(\bar{g}(\mathbf{x})-f(\mathbf{x})\right)^2\right]}_{\text{bias}}+\underbrace{\mathbb{E}_{\epsilon,\mathbf{x}}\left[\left(\epsilon(\mathbf{x})\right)^2\right]}_{\sigma^2}$$

↑ deterministic noise

↑ stochastic noise

-- Given the hypothesis set the **deterministic noise** is fixed.
-- There is nothing we can do about the **stochastic noise** as well.
-- The algorithm will try to fit both types of noise.

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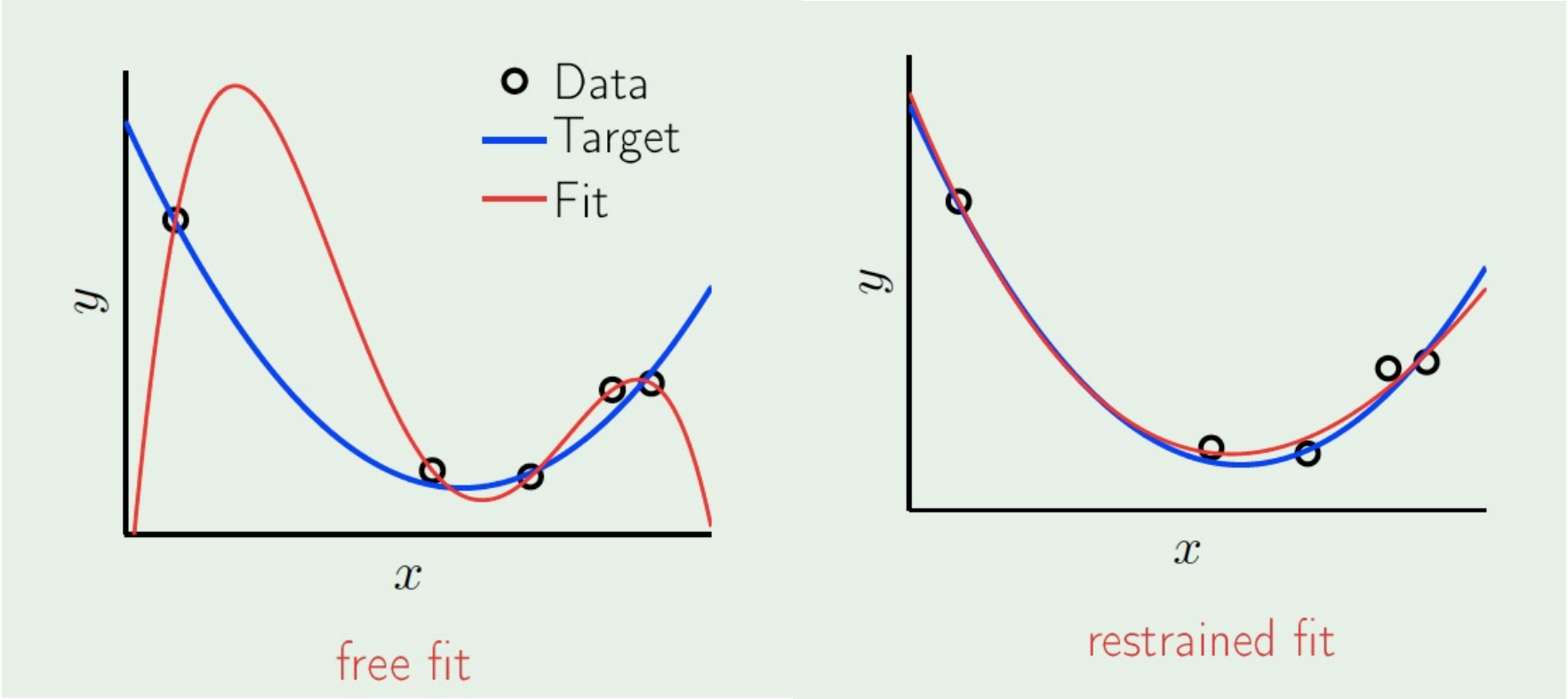
Two cures

Regularization: Putting the brakes

Validation: Checking the bottom line

Check E_{out} instead of E_{in}

Putting the brakes



The fit is 4th order polynomial.

Fit a 4th order polynomial (red curve) by restricting the curve not to pass through the points.